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Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 7	
TITLE : Inheritance and Interfaces	

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction.

Programme

```

1  interface Vehicle
2  {
3
4  void start();
5  }
6
7  class Car implements Vehicle
8  {
9
10     void drive()
11     {
12         System.out.println("Car is Running");
13     }
14
15     public void start()
16     {
17         System.out.println("Car Started");
18     }
19 }
20
21 class ElectricCar extends Car
22 {
23
24     void charge()
25     {
26         System.out.println("Electric Car is Charging");
27     }
28 }
29
30 public class Demo
31 {
32
33     public static void main(String[] args)
34     {
35
36         ElectricCar car = new ElectricCar();
37     }
38 }

```

Output



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  CODE REFERENCE LOG
piyushrohokale@Piyushs-MacBook-Air assignment06 % javac Demo.java && java Demo
Car Started
Car is Running
Electric Car is Charging
piyushrohokale@Piyushs-MacBook-Air assignment06 %
```

Conclusion

The program successfully demonstrates the use of inheritance and interfaces in Java. The Car class implements the Vehicle interface by providing the implementation of the start method, while the ElectricCar class inherits the features of the Car class using inheritance. This approach promotes code reusability, abstraction, and better program organization, making Java applications easier to develop and maintain.